

ISRAEL VALLEJO MUÑOZ

Marine Biologist | MSc
Environmental & Spatial
Data Analyst

CONTACT

- Mexico City, Mexico
- +52 55 2771 8407
- israelvallejo.mz@gmail.com
- LinkedIn
- Portfolio

LANGUAGES

Spanish
Native / C2



English
Proficient / C1



Italian
Intermediate / B2



TECHNICAL SKILLS

STATISTICS & MODELLING

- CPUE estimation
- Effort standardization
- Habitat suitability modelling
- Time-series analysis
- Regression modelling

GEOSPATIAL & RS

- QGIS 3.x
- Google Earth Engine
- Sentinel-2 / Landsat-8
- NDVI · LST analysis

PROGRAMMING & DATA

- R / RStudio (Intermediate)
- Python (Beginner)
- GitHub / Git
- ggplot2 · terra · sf

FIELD METHODS

- Marine & coastal surveys
- Scientific diving (PADI/NADD)
- Lab sample analysis

CERTIFICATIONS

- PADI Open Water Diver
- NADD Scientific Diver
- GIS Applied to Natural Resources & AI Integration

Marine Biologist (MSc) · Environmental & Spatial Data Analyst
R · QGIS · Google Earth Engine · Ecological Modelling · Remote Sensing

PROFESSIONAL SUMMARY

Quantitative Marine Ecologist and Environmental Data Analyst with an MSc in Marine Biology and Aquaculture (Federico II, Naples) and over 7 years of combined research and analytical experience across Mexico, Italy, and the USA. Specialized in ecological niche modelling, fisheries stock assessment, remote sensing, and spatial analysis using R and GIS. Proven ability to translate complex datasets into clear, evidence-based insights for conservation strategy and sustainability reporting. Currently contributing to global marine enforcement and fisheries compliance analysis at Sea Shepherd Global.

PROFESSIONAL EXPERIENCE

Fisheries Data Analyst Jan 2026 – Present
Sea Shepherd Global Remote

- Analyze fisheries enforcement and catch composition datasets across 2 high-priority regions (Mediterranean Sea and African coast), supporting conservation monitoring for Sea Shepherd Global.
- Conduct CPUE estimation and effort-standardized analyses to quantify fishing pressure and stock status, translating results into actionable intelligence for international scientific teams.
- Develop spatial visualizations of fishing effort and enforcement metrics, directly informing patrol deployment and strategic reporting.

Environmental Data Analyst Jan 2025 – Present
Independent Consultant Remote

- Delivered 8 end-to-end GIS and Remote Sensing reports: species distribution modelling (Rhincodon typus, Panthera onca, Mobula birostris), deforestation in Borneo (96.3% vegetation loss in 4 years), urban heat islands (Ho Chi Minh City), Blue Carbon mapping, fishing effort in Mexican MPAs, and bigeye tuna stock assessment (MSY = 56,300 t, EPO).
- Integrated 7+ spatial data sources (GBIF, WDPA, Bio-Oracle, Sentinel-2, Landsat-8, Global Fishing Watch, CONANP) across projects using R, QGIS, and Google Earth Engine.

Research Intern — Endocrinology Laboratory Feb – Oct 2024
Università degli Studi di Napoli Federico II Naples, Italy

- Processed tissue samples from aquaculture and wild fish using spectrophotometric, chromatographic, and histological techniques to assess antioxidant profiles and reproductive physiology.
- Contributed to active research on thermal stress responses in marine species across multiple sample batches.

Teaching Assistant — Marine Biology & Aquaculture Apr – Sep 2024
Università degli Studi di Napoli Federico II Naples, Italy

- Supported 6 months of undergraduate lab instruction in marine biology and aquaculture, guiding experiments and evaluating scientific reports.

Research Assistant — Shark Ecology Jul – Aug 2018
Shark Research and Conservation Program Miami, USA

- Captured, tagged, and sampled 15+ large coastal sharks across 6-7 species (Carcharhinus leucas, Sphyrna mokarran, among others), collecting blood, biopsy, and morphometric data for ecological research.

Research Assistant — Ichthyology Jul 2017 – Jan 2018
National Fish Collection (UNAM) Mexico City, Mexico

- Processed dozens of specimens via morphometric analysis and skeletal clearing, contributing to taxonomic curation of the national ichthyological collection.

Research Assistant — Marine Ichthyology Mar 2016 – Jan 2017
Instituto de Ciencias del Mar y Limnología (UNAM) Mexico City, Mexico

- Identified dozens of demersal fish specimens to species level; extracted otoliths for age and growth studies; processed and documented ichthyological samples from the Mexican Pacific.

EDUCATION

MSc in Marine Biology and Aquaculture Oct 2024
Università degli Studi di Napoli Federico II Naples, Italy

BSc in Biology Jan 2021
Universidad Nacional Autónoma de México (UNAM) Mexico City, Mexico